

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

1. Introduction

In today's highly competitive business environment, efficient supply chain and logistics management have become critical factors in achieving operational excellence and maintaining customer satisfaction. Organizations that rely on the timely movement of goods must continuously monitor and optimize their transportation networks to reduce delivery delays, minimize logistics costs, and improve overall distribution efficiency. With increasing customer expectations for faster and more reliable deliveries, businesses are adopting data-driven approaches to evaluate shipping performance and support strategic decision-making.

The food and confectionery industry depend on an efficient distribution network to ensure that products are delivered to customers in a timely and cost-effective manner while maintaining product quality. Nassau Candy Distributor, a leading confectionery distributor, manages a large number of shipments across multiple factories, regions, transportation modes, and customer locations. The complexity of these logistics operations generates significant volumes of operational data that can be analysed to identify performance trends, optimize shipping routes, and improve supply chain efficiency.

Traditional logistics reporting methods often rely on static spreadsheets and manual analysis, making it difficult for managers to monitor shipment performance in real time or quickly identify operational bottlenecks. Modern Business Intelligence (BI) and data visualization technologies provide organizations with the ability to transform large datasets into interactive dashboards that present meaningful insights through key performance indicators (KPIs), charts, maps, and dynamic filtering capabilities. These tools enable decision-makers to explore logistics performance from multiple perspectives and make informed business decisions based on current operational data. This project, "**Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor**," focuses on developing an interactive analytics dashboard to evaluate shipment performance across the organization's distribution network. The project integrates data preprocessing, feature engineering, exploratory data analysis, and interactive visualization to analyse factory-to-customer shipping operations. The dashboard was developed using **Python**, **Pandas**, **Plotly**, and **Streamlit**, providing an intuitive platform for monitoring logistics performance and supporting operational decision-making.

The shipment dataset consists of **10,194 records**, containing information related to customer orders, order dates, shipment dates, shipping methods, manufacturing facilities, destination regions, transportation costs, and gross profit. Using these

records, the project calculates the **Order-to-Shipment Duration** by measuring the time difference between the order date and shipment date. This derived metric serves as one of the primary indicators for evaluating shipping efficiency throughout the dashboard. The dashboard includes several interactive analytical components designed to support comprehensive logistics analysis. These include key performance indicators for shipment volume, average shipment duration, transportation cost, and gross profit; route efficiency analysis for identifying the shortest shipment routes; geographic heatmaps illustrating state-level shipping performance; regional bottleneck analysis; ship mode comparisons; shipment timelines; and detailed state-level performance tables. Interactive filters allow users to analyse shipment performance by date range, geographic region, state, ship mode, and shipment duration, enabling flexible exploration of the underlying data.

The primary objective of this project is to demonstrate how Business Intelligence and data analytics techniques can be applied to improve visibility into logistics operations and support data-driven decision-making. Rather than replacing existing logistics systems, the dashboard functions as a decision-support tool that helps managers monitor operational performance, identify patterns within shipment data, and evaluate distribution efficiency across different regions and transportation methods.

Overall, this project demonstrates the practical application of data analytics in supply chain management by converting raw shipment data into meaningful business insights. Through interactive visualization and performance monitoring, the developed dashboard provides an effective platform for understanding logistics operations and identifying opportunities for continuous operational improvement.

2. DATA LOADING & UNDERSTANDING

In this project, the shipment dataset of **Nassau Candy Distributor** was loaded into the Python environment and carefully examined to gain a comprehensive understanding of its contents. The dataset contains shipment information associated with factory-to-customer deliveries, including order details, shipment dates, transportation methods, factory locations, customer destinations, product information, sales, costs, and profits. Understanding these attributes was essential for developing meaningful performance metrics and interactive visualizations.

The data loading process was performed using the **Pandas** library, while **Streamlit's caching functionality** was implemented to improve application performance by reducing repeated file-loading operations. After successfully loading the dataset, an initial exploration was conducted to examine the dataset dimensions, column names, data types, and summary statistics. This preliminary analysis provided a clear understanding of the available information and established the basis for subsequent preprocessing and dashboard development.

2.1 Data Source

The dataset used for this project was provided in **Comma-Separated Values (CSV)** format, one of the most commonly used file formats for storing structured business data. CSV files are lightweight, platform-independent, and compatible with most programming languages and data analysis tools, making them suitable for analytics projects.

The dataset represents historical shipment transactions of Nassau Candy Distributor and contains operational information related to customer orders, shipment processing, transportation modes, geographic locations, product details, and financial performance. Each row corresponds to an individual shipment transaction, while each column represents a specific business attribute describing that shipment.

The dataset serves as the primary source of information for evaluating shipping efficiency and identifying operational trends within the factory-to-customer distribution network.

2.2 Data Loading

The shipment dataset was imported into Python using the **Pandas** library. Pandas provides efficient functions for reading, managing, and analysing structured datasets. To improve dashboard performance, Streamlit's caching mechanism was used so that the dataset is loaded only once during execution unless the underlying file changes.

2.3 Dataset Overview

After loading the dataset, its overall structure was examined to verify successful import and understand the available information.

The dataset contains **10,194 shipment records**, representing customer orders fulfilled by multiple manufacturing facilities across different regions and states. The data includes operational, geographical, and financial attributes that support logistics performance analysis.

The dataset includes information related to:

- Customer order information
- Shipment dates
- Shipping methods
- Manufacturing factories
- Destination states
- Geographic regions
- Product information
- Sales values
- Shipping costs
- Gross profit
- Factory-to-customer routes

These variables provide sufficient information to analyse shipping performance from multiple business perspectives.

3. Data Cleaning:

3.1 Clean column names

The first step in the data preprocessing stage was to clean and standardize the column names of the dataset. Data collected from different sources often contains inconsistencies such as leading or trailing spaces, special characters, inconsistent capitalization, or unnecessary whitespace. These inconsistencies can lead to errors during data manipulation, filtering, aggregation, and visualization.

To ensure smooth data processing and improve code readability, all column names were cleaned before performing any further analysis.

3.2 Understanding Data Types

Understanding the data types is an essential preprocessing step because different analytical operations require different data formats. Numerical variables are used for statistical calculations, categorical variables support grouping and filtering operations, and date variables enable time-based analysis.

The Order Date and Ship Date columns were converted to the DateTime data type to facilitate chronological analysis and accurate calculation of the order-to-shipment duration. Financial variables such as Sales, Cost, and Gross Profit were stored as floating-point numbers, enabling mathematical computations and performance analysis. Variables such as Factory, Ship Mode, Product Name, Route_State, and Route_Region were stored as string (object) data types because they represent categorical information used throughout the dashboard for filtering, grouping, and visualization.

3.3 Convert Date Columns

The conversion was performed using the **pd.to_datetime()** function provided by the Pandas library. This function transforms textual date values into DateTime objects, enabling Python to recognize and process them as dates rather than ordinary strings. Once converted, the dates could be used for filtering records by date range, calculating shipment duration, sorting chronologically, and creating timeline visualizations within the dashboard.

Data Type	Examples
Integer	Units, Lead Time
Float	Sales, Cost, Gross Profit
String	Factory, Ship Mode, Product Name, Route_State
DateTime	Order Date, Ship Date

3.4 Handle missing values

In this project, the dataset was examined to identify any missing values present in each column. The **isnull()** and **sum()** functions provided by the Pandas library were used to determine the total number of missing values for every attribute.

Also dropped rows where order date and ship date was missing.

3.5 Standardize geographic fields

The geographic attributes, including **State/Province**, **Country/Region**, **Route_State**, and **Route_Region**, were standardized by removing unnecessary whitespace and converting the text into a consistent title case format. This ensured that identical locations were represented uniformly throughout the dataset.

The `astype(str)` function converts all values into string format before processing. The `str.strip()` method removes any leading or trailing whitespace, while the `str.title()` function converts each value into title case, ensuring consistent capitalization across the dataset. Applying these operations to all geographic fields creates standardized location names that can be accurately grouped, filtered, and visualized.

4. Feature Engineering

4.1 Calculate Shipping Lead Time (days)

One of the most important features created during this project was the **Shipping Lead Time**, also referred to as the **Order-to-Shipment Duration**. This metric represents the number of days between the order placement date and the shipment date. It serves as the primary Key Performance Indicator (KPI) for evaluating the efficiency of the distribution process.

The Shipping Lead Time was calculated by subtracting the **Order Date** from the **Ship Date**. Since both columns had already been converted into the `DateTime` format during data preprocessing, the calculation could be performed accurately using the Pandas library.

During data exploration, it was observed that the calculated lead times were unusually large (approximately **900–1600 days**) because the dataset contained shipment dates that were much later than the corresponding order dates. These values did not represent realistic operational shipping durations and would have significantly distorted the logistics performance analysis.

4.2 Categorize Routes by Factory to Customer Region

To analyze shipment performance across broader geographical areas, a new feature called **Route_Region** was created by combining the **Factory** and **Country/Region** fields. This feature uniquely represents the shipment path from the manufacturing location to the customer's geographical region.

The **Route_Region** feature enabled the comparison of shipping performance across different regional destinations. It was used to identify high-volume routes, evaluate regional shipment efficiency, detect geographic bottlenecks, and generate regional performance visualizations in

the Streamlit dashboard. This categorization also supported strategic decision-making by highlighting regions that required operational improvements. Eg. Wicked Choccy's → Interior, Wicked Choccy's → Atlantic, Lot's O' Nuts → Gulf, Wicked Choccy's → Gulf etc.

4.3 Categorize Routes by Factory to Customer State

In addition to regional analysis, a more detailed route categorization was created at the state level. The **Route_State** feature combines the **Factory** and **State/Province** columns to uniquely identify shipments from each factory to individual customer states.

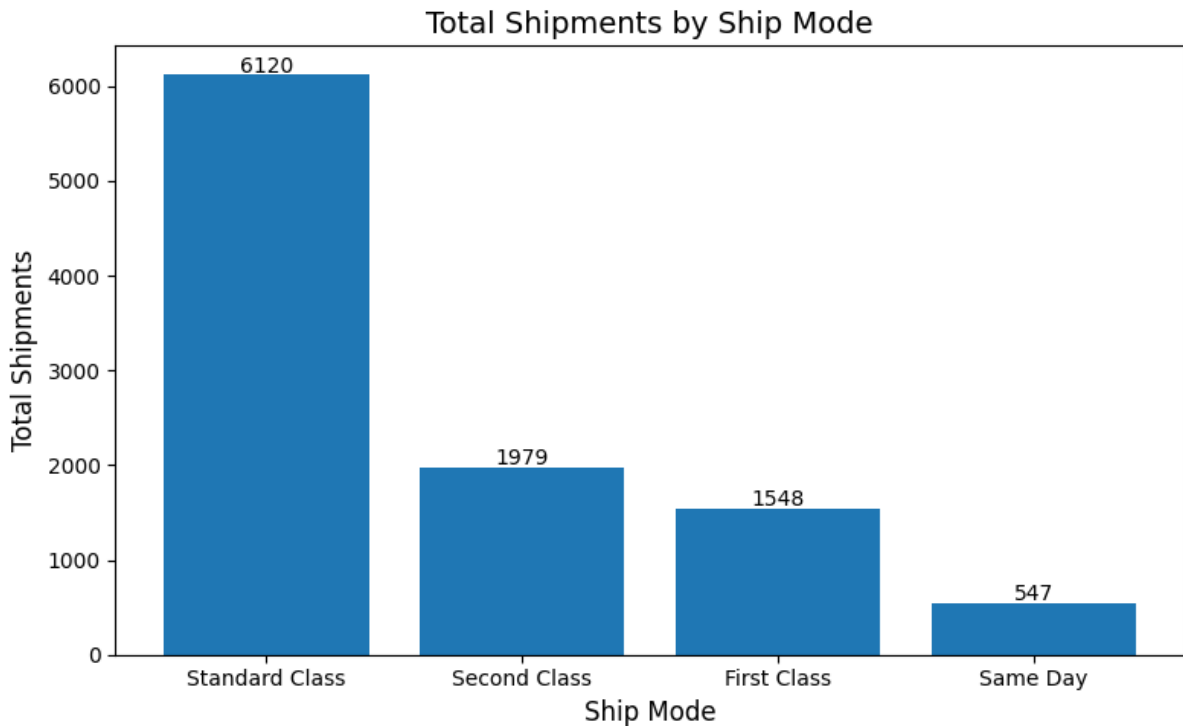
The **Route_State** feature enabled granular analysis of shipment performance for individual states. This categorization was particularly useful for identifying the fastest and slowest shipping routes, analyzing state-wise shipment volumes, evaluating average shipping duration by destination, and detecting localized distribution bottlenecks.

4.4 Group Shipments by Ship Mode

Different transportation methods have varying impacts on delivery speed, operational cost, and overall logistics efficiency. Therefore, shipment records were grouped according to the **Ship Mode** attribute to compare the performance of each transportation method.

Grouping shipments by **Ship Mode** enabled comparison of average shipment duration and shipment volume across different transportation methods. This analysis helped identify which shipping modes delivered products more efficiently and which methods handled the highest number of shipments. The aggregated data was presented in the dashboard using interactive bar charts, allowing users to compare transportation performance and make informed logistics decisions.

	Ship Mode	Total_Shipments
3	Standard Class	6120
2	Second Class	1979
0	First Class	1548
1	Same Day	547



The shipment distribution by ship mode was analysed to understand the organization's transportation preferences and the utilization of different delivery services. The analysis revealed that Standard Class was the most frequently used shipping mode, accounting for 6,120 shipments, which represents approximately 60% of all shipments. This indicates that the organization primarily relies on Standard Class as its default transportation option, likely due to its balance between operational cost and delivery efficiency.

Second Class was the second most commonly used shipping mode with 1,979 shipments (19.4%), followed by First Class with 1,548 shipments (15.2%). These shipping modes are typically selected when faster delivery is required while maintaining reasonable transportation costs.

The Same Day shipping service recorded the lowest utilization with 547 shipments (5.4%), indicating that urgent deliveries represent only a small proportion of the overall logistics operations. This suggests that most customer orders do not require immediate delivery, allowing the company to optimize shipping costs through standard transportation methods.

Key Findings

- **Standard Class: 6,120 shipments (60.0%)** – Highest utilization and primary shipping method.
- **Second Class: 1,979 shipments (19.4%)** – Second most frequently used shipping option.
- **First Class: 1,548 shipments (15.2%)** – Used for moderately urgent deliveries.

- **Same Day: 547 shipments (5.4%)** – Least utilized shipping mode, reserved for urgent customer requirements.

4.5 Route Definition & Aggregation

In this project, each transportation route was defined as the movement of goods from a Factory Location to a Customer State or Customer Region. Two route categories were created during the feature engineering process:

- Factory → Customer State (Route_State)
- Factory → Customer Region (Route_Region)

These route definitions served as the basis for aggregating shipment records and calculating important logistics performance indicators.

- **Route Definition**

Each shipment was assigned to a unique route by combining the factory location with the customer destination. This created a standardized representation of the shipment path, enabling efficient grouping and comparative analysis.

- **Route Aggregation**

After defining the routes, shipment records were aggregated using the **groupby()** function in Pandas. Aggregation combines all shipments belonging to the same route and computes summary statistics that describe the overall performance of that route.

The newly created route identifiers uniquely represent the transportation path from the manufacturing facility to the customer's destination. These route fields were later used for grouping shipment records and calculating route-level performance metrics.

4.6 Total Shipments

The **Total Shipments** metric represents the number of orders transported through each route during the analysis period. This metric indicates the shipment volume handled by a particular route and helps identify the most frequently used transportation paths.

	Route_State	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
0	Lot's O' Nuts → Alabama	34	1295.21	253.07
1	Lot's O' Nuts → Alberta	16	1273.88	266.92
2	Lot's O' Nuts → Arizona	111	1306.20	253.90
3	Lot's O' Nuts → Arkansas	31	1285.35	274.53
4	Lot's O' Nuts → British Columbia	18	1273.78	307.08
5	Lot's O' Nuts → California	1125	1316.91	265.37
6	Lot's O' Nuts → Colorado	103	1347.53	252.65
7	Lot's O' Nuts → Connecticut	47	1420.55	249.39
8	Lot's O' Nuts → Delaware	43	1273.30	251.92
9	Lot's O' Nuts → District Of Columbia	8	1365.88	168.58

The results indicate that **Lot's O' Nuts → California** is the busiest shipping route, recording **1,125 shipments**, which is substantially higher than all other state-level routes. The route reported an **average lead time of 1,316.91 days** with a **lead time variability of 265.37 days**, indicating a high shipment volume accompanied by considerable variation in delivery duration.

Other major shipping routes include **Lot's O' Nuts → Arizona** with **111 shipments** and an average lead time of **1,306.20 days**, and **Lot's O' Nuts → Colorado** with **103 shipments** and an average lead time of **1,347.53 days**. Although these routes handle fewer shipments than California, they exhibit similarly high lead-time values.

Among the routes shown, **Lot's O' Nuts → Connecticut** recorded the **highest average lead time of 1,420.55 days**, while **Lot's O' Nuts → District of Columbia** recorded an average lead time of **1,365.88 days**. These routes may require further investigation to determine the underlying causes of prolonged shipment durations.

The **lowest average lead times** among the displayed routes were observed for **Lot's O' Nuts → Delaware (1,273.30 days)** and **Lot's O' Nuts → British Columbia (1,273.78 days)**. Although these values are lower than the other routes presented, they remain substantially high in the context of the source dataset.

Lead time variability ranged from **168.58 days** for **District of Columbia** to **307.08 days** for **British Columbia**. Lower variability indicates relatively consistent shipment durations, whereas higher variability suggests greater fluctuations in delivery performance across shipments.

Business Insight

The analysis shows that **California** is the primary destination in terms of shipment volume, making it a strategically important route for logistics planning and resource allocation. Routes such as **Connecticut** and **District of Columbia** exhibit comparatively higher average lead times, while **British Columbia** shows the highest variability in shipment duration, indicating less consistent delivery performance.

4.7 Regional Shipping Lead Time Analysis

	Route_Region	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
0	Lot's O' Nuts → Atlantic	1661	1326.83	255.99
1	Lot's O' Nuts → Gulf	897	1313.98	266.59
2	Lot's O' Nuts → Interior	1321	1324.20	258.41
3	Lot's O' Nuts → Pacific	1813	1317.53	266.67
4	Secret Factory → Atlantic	72	1349.25	280.72
5	Secret Factory → Gulf	37	1332.54	304.83
6	Secret Factory → Interior	45	1289.09	311.23
7	Secret Factory → Pacific	63	1307.73	276.18
8	Sugar Shack → Atlantic	18	1375.39	274.61
9	Sugar Shack → Gulf	4	1091.25	210.45

Regional shipping route analysis was performed to evaluate shipment volume, average shipping lead time, and lead-time variability across different factory-to-region routes. This analysis provides insight into regional shipment distribution, delivery consistency, and identifies routes that may require operational attention.

The results indicate that **Lot's O' Nuts → Pacific** is the busiest regional shipping route, handling **1,813 shipments**, followed by **Lot's O' Nuts → Atlantic** with **1,661 shipments**, **Lot's O' Nuts → Interior** with **1,321 shipments**, and **Lot's O' Nuts → Gulf** with **897 shipments**. These four routes account for the majority

of the organization's shipment volume, indicating that Lot's O' Nuts serves as the primary manufacturing facility within the distribution network.

Among the major regional routes, **Lot's O' Nuts → Atlantic** recorded the **highest average lead time of 1,326.83 days**, while **Lot's O' Nuts → Gulf** recorded the **lowest average lead time of 1,313.98 days**. Although differences between the major regions are relatively small, they demonstrate slight variations in shipping performance across geographical areas.

The analysis also includes routes originating from **Secret Factory** and **Sugar Shack**. **Sugar Shack → Atlantic** recorded the highest average lead time among the displayed routes at **1,375.39 days**, whereas **Sugar Shack → Gulf** recorded the lowest average lead time of **1,091.25 days**. However, these routes handled only **18** and **4 shipments**, respectively, and therefore should be interpreted cautiously because of the limited sample size.

Lead-time variability provides an indication of delivery consistency. **Secret Factory → Interior** exhibited the highest variability (**311.23 days**), followed by **Secret Factory → Gulf (304.83 days)**, suggesting greater fluctuations in shipment duration. In contrast, **Sugar Shack → Gulf** recorded the lowest variability (**210.45 days**), indicating comparatively more consistent delivery times, although this observation is based on a very small shipment volume.

Business Insights

- **Lot's O' Nuts** is the dominant factory, supplying the majority of shipments across all four regions.
- The **Pacific region** handles the highest shipment volume (**1,813 shipments**), making it the most operationally significant regional route.
- The **Atlantic region** records the highest average lead time among the major Lot's O' Nuts routes (**1,326.83 days**), suggesting relatively longer shipment durations compared with the Gulf, Interior, and Pacific regions.
- **Secret Factory → Interior** shows the greatest lead-time variability (**311.23 days**), indicating the least consistent shipment performance among the displayed routes.
- **Sugar Shack** routes have comparatively low shipment volumes, so conclusions regarding their shipping performance should be interpreted with caution.

4.8 Average Shipping Lead Time

The Average Shipping Lead Time represents the average number of days required to transport products from the warehouse to customers after data preprocessing.

4.9 Lead Time Variability

$$\text{Lead Time Variability} = \sigma(\text{Lead Time})$$

where:

- = Standard Deviation
- Lead Time = Ship Date – Order Date (in days)

5. Efficiency Benchmarking

5.1 Top 10 Most Efficient Routes

After calculating the route-level performance metrics, all transportation routes were ranked based on their Average Shipping Lead Time. Routes with the lowest average lead time were considered the most efficient because they delivered shipments in the shortest amount of time. Ranking the routes enabled the identification of best-performing transportation corridors and provided valuable insights for improving logistics operations.

The ranking was performed by sorting all routes in ascending order of their average shipping lead time. The top-ranked routes represent the fastest and most efficient delivery paths within the distribution network.

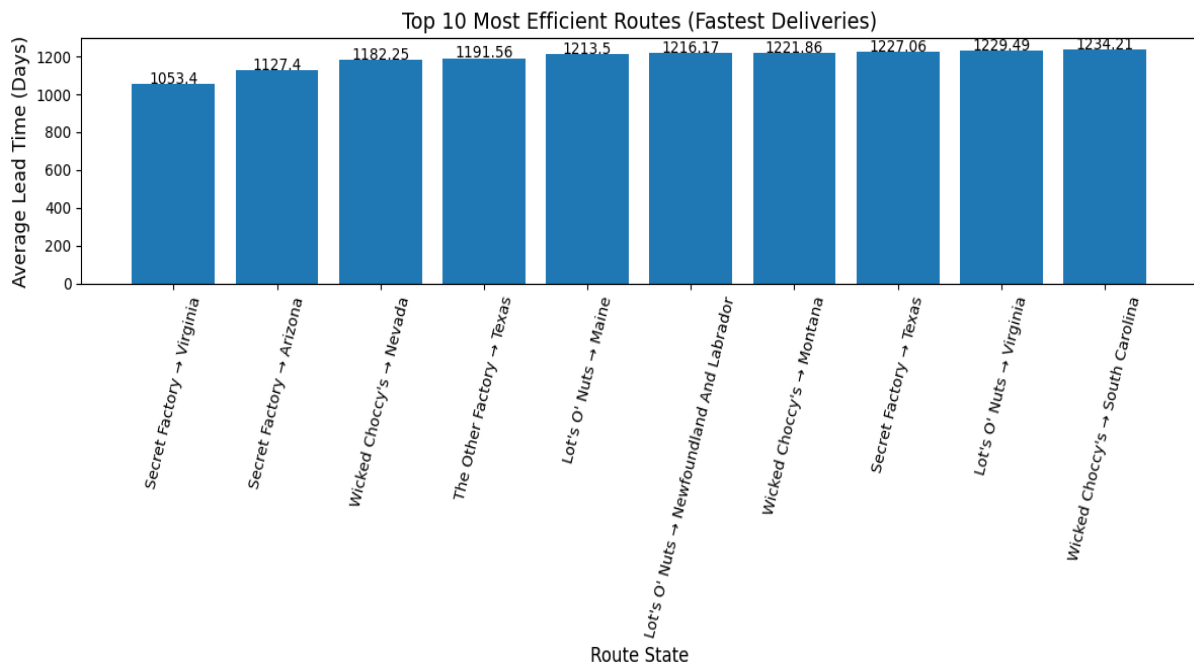
	Route_State	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
93	Secret Factory → Virginia	5	1053.40	200.02
59	Secret Factory → Arizona	5	1127.40	327.53
168	Wicked Choccy's → Nevada	12	1182.25	226.37
136	The Other Factory → Texas	9	1191.56	304.52
19	Lot's O' Nuts → Maine	6	1213.50	149.66
35	Lot's O' Nuts → Newfoundland And Labrador	6	1216.17	357.45
166	Wicked Choccy's → Montana	7	1221.86	389.90
90	Secret Factory → Texas	16	1227.06	323.91
52	Lot's O' Nuts → Virginia	109	1229.49	262.01
186	Wicked Choccy's → South Carolina	19	1234.21	240.90

The top 10 most efficient shipping routes were identified by ranking factory-to-state routes according to their **lowest average shipping lead time**. In this analysis, routes with lower average lead times are considered more efficient because they represent comparatively faster deliveries within the source dataset.

The analysis indicates that **Secret Factory → Virginia** is the most efficient route, recording the **lowest average lead time of 1,053.40 days** across **5 shipments**. This was followed by **Secret Factory → Arizona**, with an average lead time of **1,127.40 days**, and **Wicked Choccy's → Nevada**, with an average lead time of **1,182.25 days**.

The remaining routes recorded average lead times ranging from **1,191.56 days** to **1,234.21 days**. Although these routes exhibit comparatively lower lead times than other routes in the dataset, most of them processed relatively few shipments, ranging from **5 to 19 shipments**. Therefore, their performance should be interpreted cautiously because limited shipment volume may not accurately represent long-term operational efficiency.

Among the top ten routes, **Lot's O' Nuts → Virginia** is particularly noteworthy. Despite handling a considerably larger shipment volume (**109 shipments**), it maintained an average lead time of **1,229.49 days**, making it one of the most operationally significant efficient routes.



Business Insights

- **Secret Factory → Virginia** recorded the **lowest average lead time (1,053.40 days)** among all analysed routes.
- Most of the top-performing routes processed **fewer than 20 shipments**, indicating that their rankings are based on relatively limited operational activity.
- **Lot's O' Nuts → Virginia** stands out because it combines **high shipment volume (109 shipments)** with comparatively low average lead time, making it a more reliable indicator of operational efficiency.
- The average lead times for the top ten routes vary by approximately **181 days** (from **1,053.40** to **1,234.21** days), demonstrating measurable differences in shipping performance across factory-to-state routes.

5.2 The Bottom 10 Slowest Routes

	Route_State	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
37	Lot's O' Nuts → North Dakota	5	1638.20	1.64
51	Lot's O' Nuts → Vermont	5	1492.40	200.29
172	Wicked Choccy's → New Mexico	17	1488.88	226.43
65	Secret Factory → Florida	7	1482.14	196.26
15	Lot's O' Nuts → Iowa	16	1479.00	187.51
47	Lot's O' Nuts → South Dakota	9	1477.22	322.72
94	Secret Factory → Washington	11	1472.36	190.50
33	Lot's O' Nuts → New Mexico	18	1456.94	258.26

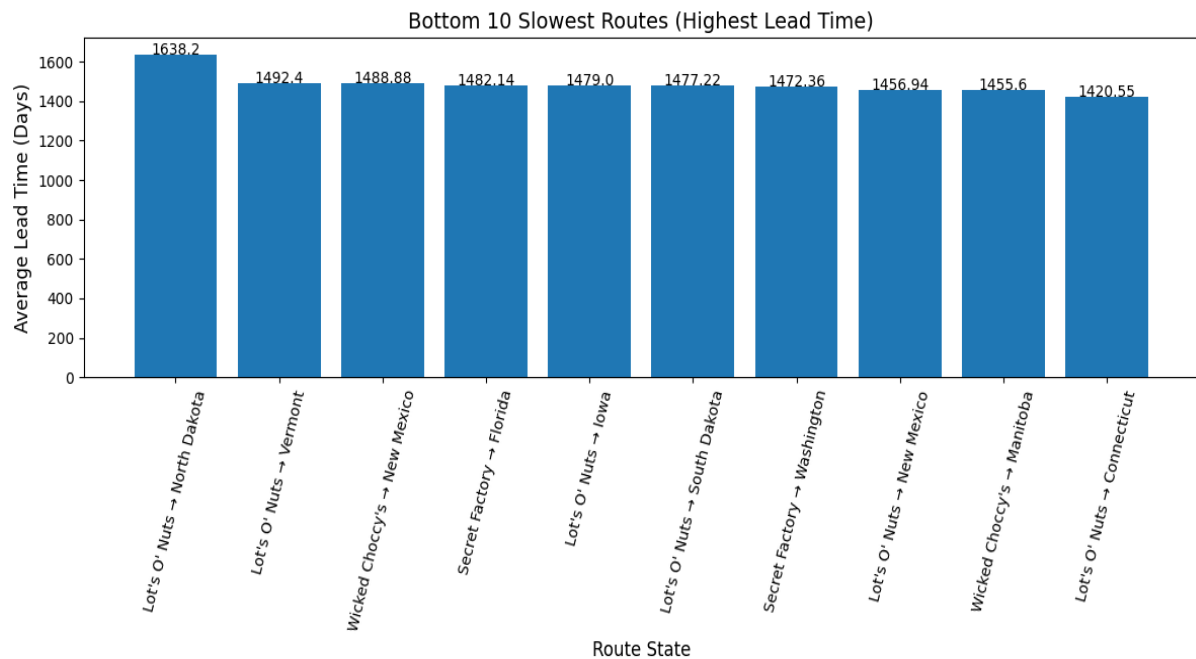
	Route_State	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
159	Wicked Choccy's → Manitoba	10	1455.60	192.90
7	Lot's O' Nuts → Connecticut	47	1420.55	249.39

The bottom 10 shipping routes were identified by ranking factory-to-state routes according to their highest average shipping lead time. These routes represent the least efficient shipping paths within the analysed dataset and highlight areas where shipment durations were comparatively longer than other routes.

The analysis shows that Lot's O' Nuts → North Dakota recorded the highest average lead time of 1,638.20 days, making it the slowest-performing route among all analysed factory-to-state combinations. This was followed by Lot's O' Nuts → Vermont, with an average lead time of 1,492.40 days, and Wicked Choccy's → New Mexico, with 1,488.88 days.

Other routes in the bottom ten include Secret Factory → Florida (1,482.14 days), Lot's O' Nuts → Iowa (1,479.00 days), Lot's O' Nuts → South Dakota (1,477.22 days), and Secret Factory → Washington (1,472.36 days). These routes consistently recorded higher average shipment durations than the remaining shipping network.

Although Lot's O' Nuts → Connecticut appears in the bottom ten with an average lead time of 1,420.55 days, it recorded the lowest average lead time within this group, indicating comparatively better performance than the other slowest routes.



Business Insights

- **Lot's O' Nuts → North Dakota** recorded the **highest average lead time (1,638.20 days)**, making it the slowest route in the dataset.
- Several routes from **Lot's O' Nuts** appear in the bottom ten, indicating that multiple destinations supplied by this factory experienced comparatively higher shipment durations.
- Routes associated with **Secret Factory** and **Wicked Choccy's** also appear among the slowest-performing routes, suggesting that longer lead times are distributed across multiple factories rather than being isolated to a single manufacturing location.
- The difference between the slowest route (**1,638.20 days**) and the least delayed route within this group (**1,420.55 days**) is approximately **218 days**, highlighting measurable variation in shipment performance across the lowest-ranked routes.

5.3 Compare Performance Across Ship Modes

	Ship Mode	Total_Shipments	Avg_Lead_Time	Lead_Time_Variability
0	First Class	1548	1338.28	265.63
1	Same Day	547	1333.44	253.81
2	Second Class	1979	1323.85	261.81
3	Standard Class	6120	1314.33	262.40

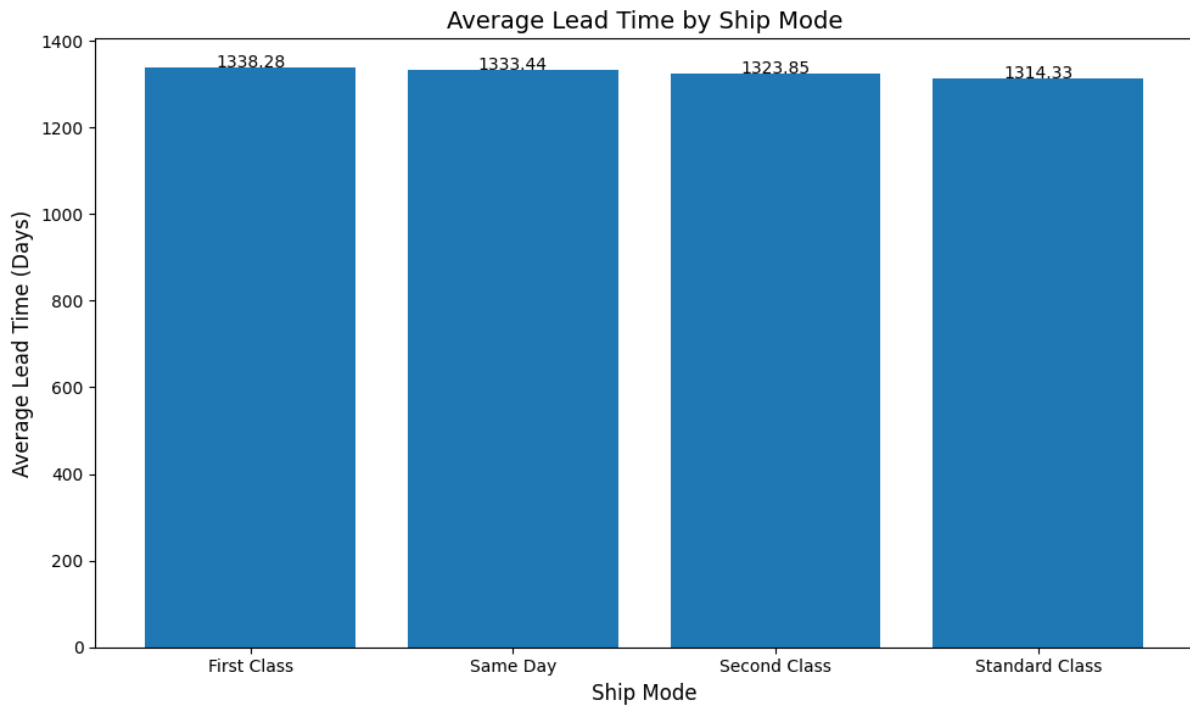
The results indicate that **Standard Class** is the most frequently used shipping method, handling **6,120 shipments**, which accounts for approximately **60%** of all shipments. It also recorded the **lowest average lead time of 1,314.33 days**, making it the fastest-performing shipping mode within the analysed dataset.

Second Class processed **1,979 shipments** with an average lead time of **1,323.85 days**, followed by **Same Day** shipping with **547 shipments** and an average lead time of **1,333.44 days**. Surprisingly, **Same Day** shipping did not achieve the lowest lead time in the dataset, suggesting that the source data does not reflect typical real-world delivery service behaviour.

First Class recorded **1,548 shipments** and the **highest average lead time of 1,338.28 days**, indicating comparatively longer shipment durations than the other shipping modes.

Lead Time Variability ranged between **253.81 days** and **265.63 days**, indicating that all shipping modes exhibit a relatively similar level of variation in shipment duration. **Same Day** shipping showed the lowest variability (**253.81 days**), suggesting comparatively more consistent delivery performance, whereas **First Class** recorded the highest variability (**265.63 days**).

Here wording like "**Within the analysed dataset, Standard Class recorded the lowest average lead time.**"

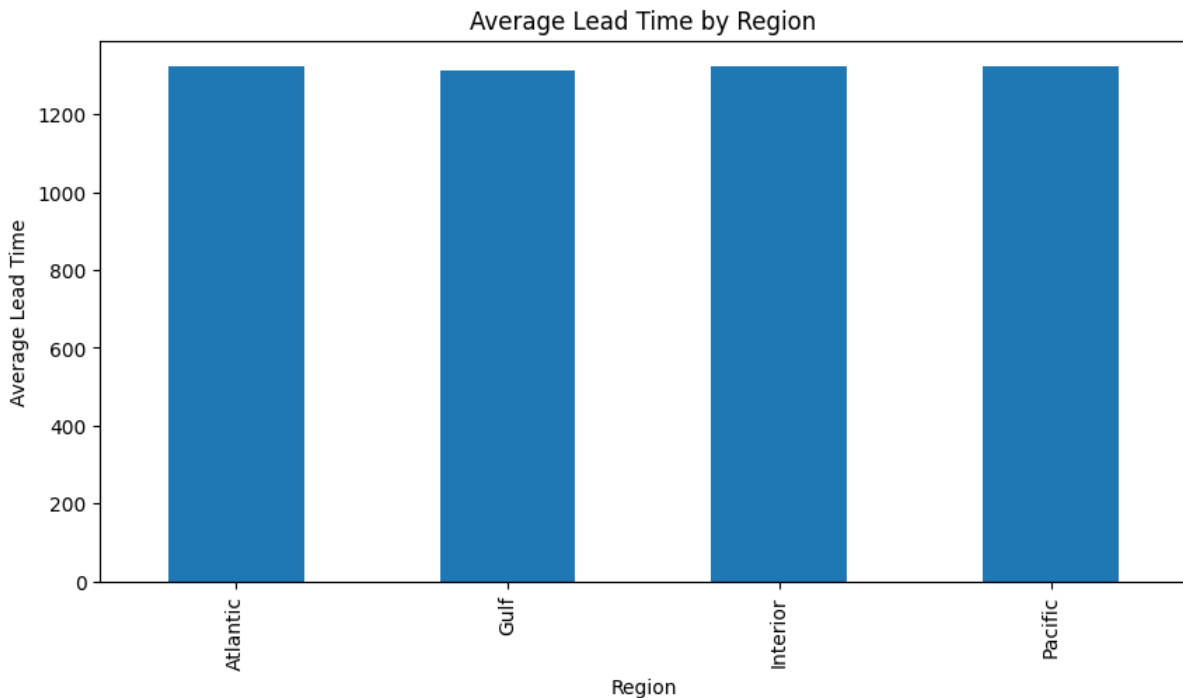


Interpretation of Results

- **Standard Class** is the dominant shipping method, processing approximately **60%** of all shipments while also recording the lowest average lead time (**1,314.33 days**).
- **Second Class** demonstrates similar operational performance with only a modest increase in average lead time compared with Standard Class.
- **Same Day** shipping processes the smallest number of shipments (**547**) and has the **lowest lead-time variability**, indicating relatively consistent delivery durations within the dataset.
- **First Class** records the highest average lead time and the highest variability among the four shipping modes, indicating comparatively slower and less consistent performance in the source data.

6. Geographic Bottleneck Analysis

6.1 High average lead time by Region Level Analysis



The results indicate that the **Atlantic** region recorded the **highest average lead time of 1,326.83 days**, followed closely by the **Interior** region with **1,324.20 days**. The **Pacific** region reported an average lead time of **1,317.53 days**, while the **Gulf** region recorded the **lowest average lead time of 1,313.98 days**.

Although differences exist between the regions, the variation in average lead time is relatively small. The difference between the highest-performing region (Gulf) and the lowest-performing region (Atlantic) is approximately **13 days**, indicating that shipping performance is broadly consistent across all geographic regions in the dataset.

Overall, the regional analysis indicates that shipment performance is **fairly consistent across all four customer regions**, with only minor differences in average lead time. While the **Atlantic** region recorded the highest average lead time and the **Gulf** region the lowest, the limited variation between regions suggests that regional location alone does not appear to be the primary factor influencing shipment duration within the analysed dataset.

❖ Worst Performing Region

Interior Region

The **Interior region** is the **worst-performing region** in terms of average shipping lead time.

Supporting Evidence

- **Average Lead Time: 1,323.09 days** (highest among all regions)
- **Total Shipments: 2,335 shipments**

Although the Interior region is not the highest-volume region, it records the **longest average shipment duration**, indicating comparatively slower shipping performance within the dataset.

❖ Identify Critical Regions :

$$\text{Bottleneck Score} = \text{Avg_Lead_Time} \times \text{Total_Shipments}$$

	Region	Avg_Lead_Time	Total_Shipments	Bottleneck_Score
3	Pacific	1322.194897	3253	4301100.0
0	Atlantic	1322.745144	2986	3949717.0
2	Interior	1323.091221	2335	3089418.0
1	Gulf	1311.374691	1620	2124427.0

Executive Summary Statement

Critical Region: Pacific

- **Average Lead Time: 1,322.19 days**
- **Total Shipments: 3,253**
- **Bottleneck Score: 4,301,100 (Highest)**
- **Reason: Although its average lead time is similar to the other regions, the Pacific region processes the largest shipment volume. As a result, delays in this region have the greatest overall operational impact, making it the highest priority for logistics improvement.**

Interpretation Guide

It's useful to distinguish between two different perspectives in your report:

- **Worst-performing region by lead time: Interior (highest average lead time: 1,323.09 days).**
- **Most critical region by operational impact: Pacific (highest Bottleneck Score due to the combination of high shipment volume and high average lead time).**

Executive Summary Statement

Critical Region: Pacific

- Average Lead Time: 1,322.19 days
- Total Shipments: 3,253
- Bottleneck Score: 4,301,100 (Highest)
- Reason: Although its average lead time is similar to the other regions, the Pacific region processes the largest shipment volume. As a result, delays in this region have the greatest overall operational impact, making it the highest priority for logistics improvement.

Interpretation Guide

It's useful to distinguish between two different perspectives in your report:

- Worst-performing region by lead time: Interior (highest average lead time: 1,323.09 days).
- Most critical region by operational impact: Pacific (highest Bottleneck Score due to the combination of high shipment volume and high average lead time).

Final Ranking of Critical Regions

1. Pacific – Highest operational impact (highest bottleneck score)
2. Atlantic – High operational priority
3. Interior – Highest delivery delay (worst lead time)
4. Gulf – Best-performing region

6.2 High shipment volume + poor performance and congestion prone states:

To identify congestion-prone states within the distribution network, a Congestion Score was developed by combining the Average Shipping Lead Time and the Total Shipments for each state. The Congestion Score was calculated by first computing a Congestion Index as the product of Average Lead Time and Total Shipments. The index was then normalized to a 0–100 scale, enabling meaningful comparison of congestion levels across all states.

The Congestion Score was calculated using the following formula:

$$\text{Congestion Score} = \frac{\text{Average Lead Time} \times \text{Total Shipments}}{\max(\text{Average Lead Time} \times \text{Total Shipments})} \times 100$$

	State/Province	Total_Shipments	Avg_Lead_Time	Congestion_Score
5	California	2001	1318.427286	100.00
34	New York	1128	1324.493794	56.63
51	Texas	985	1310.305584	48.92
43	Pennsylvania	587	1324.199319	29.46
55	Washington	506	1360.662055	26.10
13	Illinois	492	1323.150407	24.68
39	Ohio	469	1325.733475	23.57
10	Florida	383	1311.396867	19.04
36	North Carolina	249	1334.883534	12.60
23	Michigan	255	1300.560784	12.57

The analysis identified **California** as the most congestion-prone state with a **Congestion Score of 100**, substantially higher than all other states in the distribution network. This indicates that California experiences the greatest overall logistics workload and should be considered the highest-priority state for supply chain optimization.

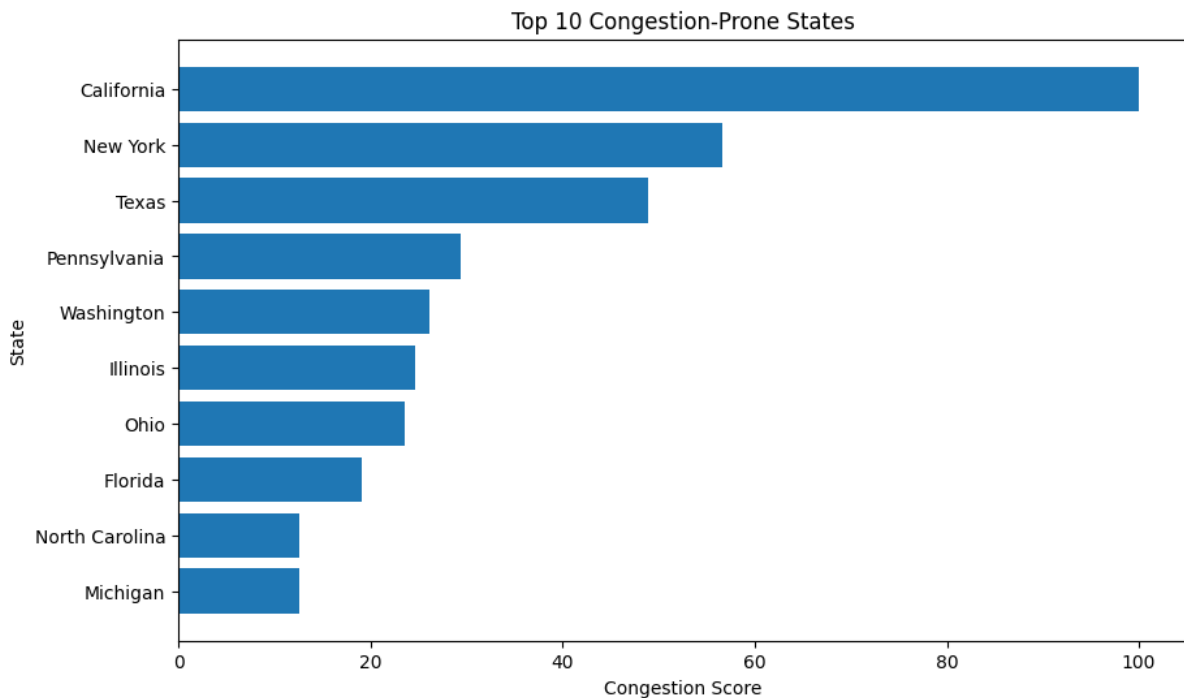
New York ranked second with a congestion score of approximately **57**, followed by **Texas** with a score of approximately **49**. These states also represent major distribution destinations and contribute significantly to the overall transportation workload.

The remaining states, including **Pennsylvania, Washington, Illinois, Ohio, Florida, North Carolina, and Michigan**, recorded comparatively lower congestion scores ranging from approximately **13 to 30**. Although these states experience lower operational pressure than California, they remain important distribution locations that should continue to be monitored for changes in shipment demand.

Overall, the analysis demonstrates that logistics congestion is concentrated in a small number of high-volume destination states, while the majority of states experience comparatively moderate operational demand.

Key Findings

- **California** is the **most congestion-prone state**, with a congestion score of **100**, making it the highest operational priority.
- **New York** and **Texas** rank second and third, indicating significant transportation demand in these states.
- The large difference between California and the remaining states suggests that logistics activity is concentrated in a few major destination markets.
- Focusing operational improvements on **California**, followed by **New York** and **Texas**, is likely to provide the greatest benefits for network efficiency.
- Lower-ranked states currently exhibit comparatively lower congestion and require routine operational monitoring rather than immediate intervention.



6.3 Regional Bottleneck Classification

To identify regions requiring operational attention, a rule-based classification model was developed using two performance indicators: **Average Shipping Lead Time** and **Total Shipments**. Regions were categorized based on whether their average lead time exceeded the overall dataset average and whether their shipment volume exceeded the average regional shipment volume.

The bottleneck score for each region was calculated as:

$$\text{Bottleneck Score} = \text{Average Lead Time} \times \text{Total Shipments}$$

	Region	Avg_Lead_Time	Total_Shipments	Bottleneck_Score	Category
0	Atlantic	1322.745144	2986	3949717.0	Critical Bottleneck
1	Gulf	1311.374691	1620	2124427.0	Efficient
2	Interior	1323.091221	2335	3089418.0	Slow but Low Volume
3	Pacific	1322.194897	3253	4301100.0	Critical Bottleneck

A Regional Bottleneck Analysis was conducted to identify regions with the greatest operational impact by combining **Average Lead Time** and **Total Shipments** into a single **Bottleneck**

Score. Regions with higher scores represent areas where shipment delays affect a larger number of orders and therefore require higher operational priority.

The analysis identified the **Pacific** region as the most critical bottleneck, recording the **highest Bottleneck Score of 4,301,100** while processing **3,253 shipments** with an average lead time of **1,322.19 days**. This indicates that even though its average lead time is similar to the other regions, the large shipment volume significantly increases its operational impact.

The **Atlantic** region ranked second with a **Bottleneck Score of 3,949,717**, processing **2,986 shipments** and recording an average lead time of **1,322.75 days**. Together, the Pacific and Atlantic regions account for the largest proportion of shipments in the distribution network and therefore represent the highest operational priorities.

The **Interior** region recorded the **highest average lead time (1,323.09 days)** among all regions but handled fewer shipments (**2,335**). Consequently, its Bottleneck Score (**3,089,418**) is lower than those of the Pacific and Atlantic regions. This region has been classified as "**Slow but Low Volume,**" indicating that although shipment durations are comparatively longer, the operational impact is moderated by its lower shipment volume.

The **Gulf** region achieved the **lowest average lead time (1,311.37 days)** and the **lowest Bottleneck Score (2,124,427)** while processing **1,620 shipments**. As a result, it has been classified as "**Efficient,**" reflecting comparatively better shipping performance within the dataset.

Key Findings

- **Pacific** is the **most critical bottleneck region**, with the highest Bottleneck Score (**4,301,100**) driven by its large shipment volume (**3,253 shipments**).
- **Atlantic** is the **second most critical region**, with a Bottleneck Score of **3,949,717**, also reflecting high shipment demand.
- **Interior** has the **highest average lead time (1,323.09 days)** but a lower operational impact because of its smaller shipment volume.
- **Gulf** is the **best-performing region**, recording the lowest average lead time (**1,311.37 days**) and the lowest Bottleneck Score (**2,124,427**).

Management Recommendation

Based on the bottleneck analysis, management should prioritize:

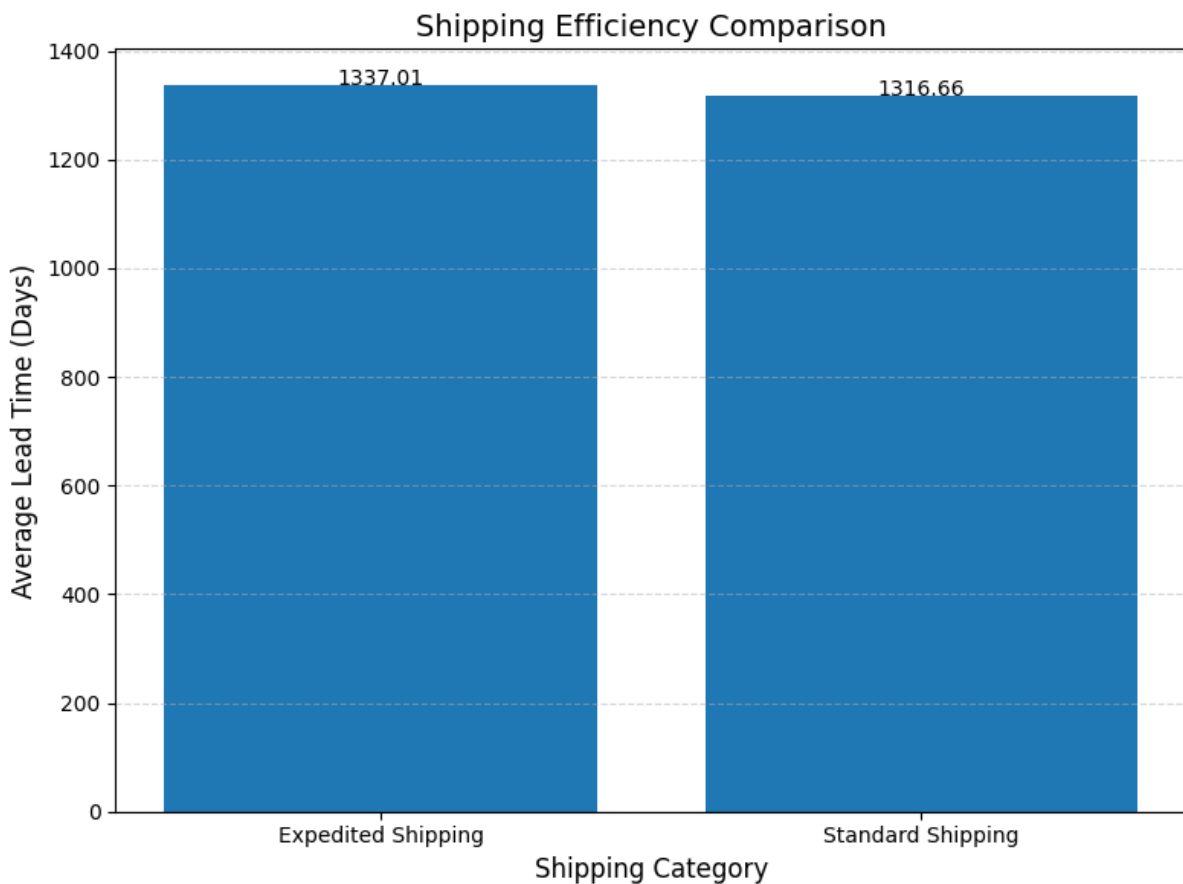
1. **Pacific Region** – Highest priority due to the greatest overall operational impact.
2. **Atlantic Region** – Second priority because of its high shipment volume and bottleneck score.
3. **Interior Region** – Focus on reducing lead times rather than increasing capacity.
4. **Gulf Region** – Maintain current operational practices and use this region as a benchmark for improving logistics performance in other regions.

7. Ship Mode Performance Analysis

7.1 Compare shipping efficiency by:

- Standard shipping
- Expedited shipping

	Shipping_Category	Total_Shipments	Avg_Lead_Time	Avg_Cost	Lead_Time_Variability
0	Expedited Shipping	2095	1337.01	4.64	262.55
1	Standard Shipping	8099	1316.66	4.77	262.27



The comparison suggests that there is no significant operational advantage associated with the expedited shipping category in this dataset. Instead, Standard Shipping recorded a marginally lower average lead time while also handling the majority of shipments (as observed in the earlier shipping mode analysis).

From an operational perspective, this indicates that the organization's standard transportation process is performing consistently despite its higher shipment volume. Since expedited shipping services did not demonstrate shorter average lead times in the available data, management should review whether premium shipping services are delivering the expected operational value.

However, because the lead-time values are derived directly from the original dataset and represent unusually long durations, these findings should be interpreted as relative comparisons within the dataset rather than evidence that Standard Shipping is inherently faster than expedited services.

Key Findings

- Standard Shipping recorded the lowest average lead time (1,316.66 days).
- Expedited Shipping recorded an average lead time of 1,337.01 days.
- The difference between the two shipping categories is approximately 20.35 days, indicating only a modest variation in performance.
- The analysis suggests that the standard shipping process performs consistently despite processing a larger share of shipments.
- Further investigation into expedited shipment handling may help determine why premium shipping services do not exhibit lower lead times within the dataset.

7.2 Cost–Time Trade-off Analysis:

1. Standard Class

- Processed the **highest number of shipments (6,120)**, indicating it is the primary shipping mode.
- Recorded the **lowest average lead time (1,314.33 days)** among all shipping modes.
- Average shipping cost was **4.75**, which is moderate compared to the other shipping modes.
- Demonstrates the **best overall balance** between shipment volume, shipping cost, and lead time within the dataset.

2. Second Class

- Handled **1,979 shipments**, making it the second most frequently used shipping mode.
- Recorded an **average lead time of 1,323.85 days**.
- Had the **highest average shipping cost (4.83)** among all shipping modes.
- Despite the higher shipping cost, it did **not** achieve the lowest lead time.

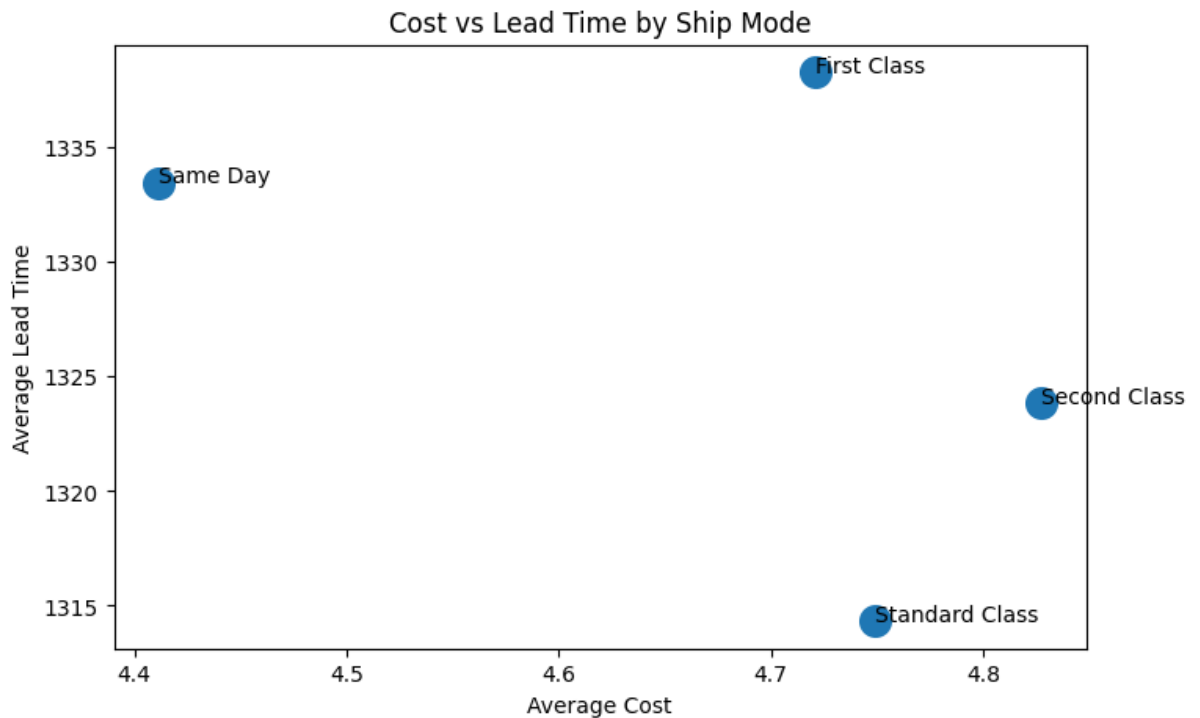
3. First Class

- Processed **1,548 shipments**.
- Recorded the **highest average lead time (1,338.28 days)**.
- Average shipping cost was **4.72**, similar to Standard Class.
- Indicates comparatively lower shipping efficiency within the analysed dataset.

4. Same Day

- Processed the **fewest shipments (547)**.
- Recorded an **average lead time of 1,333.44 days**.

- Had the **lowest average shipping cost (4.41)**.
- Despite being labelled as "Same Day," it did not record the lowest lead time in the dataset.



Key Findings

- **Most Efficient Shipping Mode:** Standard Class
- **Highest Shipping Cost:** Second Class (4.83)
- **Lowest Shipping Cost:** Same Day (4.41)
- **Highest Average Lead Time:** First Class (1,338.28 days)
- **Lowest Average Lead Time:** Standard Class (1,314.33 days)
- **Highest Shipment Volume:** Standard Class (6,120 shipments)

Note: The lead-time values are calculated directly from the original **Order Date** and **Ship Date** fields in the dataset and have been intentionally preserved. Therefore, the results should be interpreted as **comparative performance within the dataset**, rather than as representative of actual real-world delivery times.

7.3 Lead Time Vs Cost Ratio

our **Lead Time vs Cost Ratio** analysis is suitable for your dissertation, but you should clearly define the metric before discussing the results.

Assuming you calculated:

$$\text{Lead Time per Cost} = \frac{\text{Average Lead Time}}{\text{Average Cost}}$$

Lead Time VS COST RATIO

	Ship Mode	Avg_Lead_Time	Avg_Cost	Total_Shipments	Lead Time_per_Cost
1	Same Day	1333.442413	4.411225	547	302.283936
0	First Class	1338.275840	4.721124	1548	283.465512
3	Standard Class	1314.334641	4.749328	6120	276.741156
2	Second Class	1323.845376	4.827539	1979	274.227786

Same Day

- Recorded the highest Efficiency per Cost ratio (302.28).
- Average Lead Time: 1333.44 days
- Average Cost: 4.41
- Processed 547 shipments.
- The high ratio is mainly due to its relatively low average shipping cost.

First Class

- Efficiency per Cost ratio: 283.47
- Average Lead Time: 1338.28 days
- Average Cost: 4.72
- Total Shipments: 1,548
- Ranked second based on the calculated ratio.

Standard Class

- Efficiency per Cost ratio: 276.74
- Lowest Average Lead Time: 1314.33 days
- Average Cost: 4.75
- Highest shipment volume (6,120 shipments).
- Represents the primary shipping mode in the distribution network.

Second Class

- Efficiency per Cost ratio: 274.23

- Average Lead Time: 1323.85 days
- Highest Average Cost: 4.83
- Total Shipments: 1,979
- Recorded the lowest ratio among the four shipping modes.

Business Interpretation

The Efficiency per Cost ratio measures the relationship between shipping lead time and transportation cost for each shipping mode.

Based on the calculated values:

- Same Day recorded the highest ratio (302.28) because it combines a relatively low shipping cost with the observed average lead time.
- Standard Class maintained the lowest average lead time while processing the largest number of shipments, demonstrating stable operational performance.
- First Class and Second Class produced intermediate ratio values, indicating similar cost-to-lead-time relationships.
- The relatively small variation in ratios (274.23–302.28) suggests that the cost and lead-time relationship is broadly similar across shipping modes.

Key Findings

- Highest Efficiency per Cost Ratio: Same Day (302.28)
- Lowest Efficiency per Cost Ratio: Second Class (274.23)
- Lowest Average Lead Time: Standard Class (1314.33 days)
- Highest Average Cost: Second Class (4.83)
- Highest Shipment Volume: Standard Class (6,120 shipments)

8. Key Performance Indicators (KPIs)

1. KPI 1 Shipping Lead Time

Formula

$\text{Shipping Lead Time} = \text{Ship Date} - \text{Order Date}$
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where:

- **Order Date** = Date when the customer placed the order.
- **Ship Date** = Date when the shipment was dispatched.
- **Shipping Lead Time** = Total number of calendar days between the two dates.

Key Findings

- Shipping Lead Time is calculated directly from the Order Date and Ship Date.
- The Lead Time and Original_Lead_Time columns currently contain identical values.
- The minimum observed lead time is approximately 909 days.
- The KPI provides the basis for route efficiency, regional bottleneck, and shipping mode analyses.

2. KPI 2 Average Lead Time

Formula

$$\text{Average Lead Time} = \frac{\sum \text{Lead Time}}{\text{Total Shipments}}$$

where:

- **Lead Time** = Number of days between Order Date and Ship Date.
- **Total Shipments** = Total number of shipments for a given route.

Average Lead Time provides a route-level measure of shipping performance by summarizing the average shipment duration for each factory-to-customer route.

The results show noticeable variation across the **196 factory-to-state routes**. For example:

- **Wicked Choccy's → Virginia** recorded an average lead time of **1,238.88 days**, indicating comparatively lower shipment duration within the dataset.
- **Lot's O' Nuts → Alberta** recorded **1,273.88 days**, which is also among the lower average lead times.
- In contrast, **Wicked Choccy's → West Virginia** recorded the **highest average lead time of 1,639.00 days**, indicating the longest shipment duration among the listed routes.

3. KPI 3 Route Volume

Route Volume measures the total number of shipments handled by each Factory-to-Customer route during the analysis period. It reflects the level of logistics activity associated with each route and is used to identify both high-demand and low-demand shipping corridors.

The KPI was calculated by counting the total number of shipment records associated with each **Route_State**.

$$\text{Route Volume} = \text{Count of Shipments for each Route}$$

The analysis identified substantial variation in shipment volume across the **196 analysed routes**. High-volume routes, such as **Wicked Choccy's → Washington (221 shipments)** and **Lot's O' Nuts → Arizona (111 shipments)**, represent key distribution corridors that require continuous monitoring and efficient resource allocation. Conversely, routes with very low shipment volumes, such as **Wicked Choccy's → West Virginia (2 shipments)**, contribute minimally to overall logistics operations and may require less operational attention.

Overall, Route Volume provides valuable insight into shipment demand and serves as a supporting KPI for evaluating route efficiency, bottleneck analysis, and transportation planning.

KPI 4 Delay Frequency

Delay Frequency measures the percentage of shipments on each Factory-to-Customer route that exceed the predefined lead-time threshold. This KPI provides insight into the reliability of shipping operations by identifying routes where shipment delays occur most frequently.

The Delay Frequency was calculated using the following equation:

$$\text{Delay Frequency (\%)} = (\text{Number of Delayed Shipments} / \text{Total Shipments}) \times 100$$

The analysis revealed substantial variation across the 196 analysed routes. While some routes exhibited relatively low delay frequencies, others recorded considerably higher values, indicating recurring shipment delays. Routes with high delay frequencies—particularly those with substantial shipment volumes—should be prioritized for operational review to improve shipping reliability and overall logistics performance.

KPI 5 Root Efficiency Score

$$\text{Efficiency Score} = 1 / \text{Average Lead Time}$$

The analysis identified noticeable differences in efficiency across the **196 analysed routes**. Routes with higher efficiency scores demonstrated comparatively shorter shipment durations and stronger operational performance, whereas routes with lower scores experienced longer average lead times and may require further operational investigation.

9. Streamlit Dashboard Analysis

9.1 Dashboard Overview

A web-based interactive dashboard was developed using **Python, Streamlit, and Plotly** to analyze the efficiency of factory-to-customer shipping routes for Nassau Candy Distributor. The dashboard provides logistics managers with an integrated view of shipment performance, regional bottlenecks, ship mode efficiency, and route-level trends. Users can dynamically filter the data by date range, region, state, ship mode, and lead-time threshold to support operational decision-making. The dashboard layout, filters, KPI cards, charts, and tables are shown throughout the uploaded dashboard pages.

9.2 Dashboard Components

The dashboard consists of the following analytical sections:

- KPI summary cards
- Route Efficiency Overview
- Geographic Shipping Heat Map
- Regional Bottleneck Analysis
- Ship Mode Comparison
- Cost vs. Lead Time Analysis
- Route Drill-Down
- State-Level Performance
- Recent Shipment Timeline
- Interactive Filtered Dataset

These components enable users to monitor shipment efficiency at multiple levels, from individual routes to regional and national performance.

KPI	Value
Total Orders	10,194
Average Shipping Lead Time	1320.84 Days
Average Transportation Cost	\$4.74
Average Gross Profit	\$9.17
Total Factory-to-Customer Routes	196
States Covered	59
Delay Percentage	100.0%

1. Route Efficiency Overview

The **Route Efficiency Overview** provides a high-level summary of the overall shipping performance of the Nassau Candy Distributor logistics network. It presents key operational metrics, including Total Orders, Average Shipping Lead Time, Average Transportation Cost, Average Gross Profit, Total Factory-to-Customer Routes, States Covered, and Delay Percentage. These KPI cards enable stakeholders to quickly assess the scale and efficiency of the distribution network before exploring detailed analyses.

The dashboard reports **10,194 shipment records**, covering **196 factory-to-customer routes** across **59 states and provinces**. The overall average shipping lead time is **1320.84 days**, with an average transportation cost of **\$4.74** and an average gross profit of **\$9.17** per shipment. These summary indicators serve as baseline performance measures for monitoring logistics operations and evaluating route efficiency over time.

2. Geographic Shipping Heat Map

The **Geographic Shipping Heat Map** visualizes the average shipping lead time across all states and provinces served by the distributor. The choropleth map enables rapid identification of geographic areas experiencing longer shipment durations and highlights regional differences in delivery performance.

States displayed with darker shades represent higher average lead times and therefore indicate potential logistics bottlenecks, while lighter shades correspond to relatively faster shipping performance. This visualization allows logistics managers to identify geographic regions requiring operational improvements, carrier optimization, or warehouse network adjustments. The accompanying state-level performance table provides shipment counts and average lead times for each state to support detailed regional comparisons.

3. Regional Bottleneck Analysis

The **Regional Bottleneck Analysis** evaluates shipping performance across the four primary distribution regions: **Pacific, Atlantic, Interior, and Gulf**. A Bottleneck Score was calculated by combining shipment volume with average lead time, enabling regions to be classified according to their operational performance.

Based on the analysis:

- **Pacific Region** recorded the highest bottleneck score and was classified as a **Critical Bottleneck**.
- **Atlantic Region** also exhibited significant congestion and was categorized as a **Critical Bottleneck**.
- **Interior Region** was identified as **Slow but Low Volume**, indicating operational delays despite relatively fewer shipments.
- **Gulf Region** demonstrated the lowest bottleneck score and was classified as the **most efficient region**.

This regional assessment enables management to prioritize operational improvement efforts where logistics congestion has the greatest business impact.

4. Ship Mode Comparison

The **Ship Mode Comparison** analyzes the performance of the four shipping methods—**Standard Class, Second Class, First Class, and Same Day**—by comparing average lead time, shipment volume, transportation cost, and operational efficiency.

The analysis indicates that **Standard Class** handled approximately **60% of all shipments**, making it the most frequently used shipping mode. It also recorded the **lowest average lead time (1314.33 days)** among all shipping methods and was therefore identified as the best overall performing shipping mode. Conversely, **Same Day** accounted for only **5.37% of shipments**, indicating limited operational utilization despite its specialized service offering.

This comparison assists decision-makers in evaluating shipping mode utilization, service efficiency, and cost effectiveness for future logistics planning.

5. Cost vs. Lead Time Analysis

The **Cost vs. Lead Time Analysis** investigates the relationship between transportation cost and shipping performance across different shipping modes. A bubble chart compares average transportation cost against average lead time while also incorporating shipment volume.

An **Efficiency per Cost** metric was calculated to evaluate the operational value generated for every unit of transportation cost. According to the dashboard, **Same Day shipping achieved the highest Efficiency per Cost value (302.367)** due to its comparatively low transportation cost, while **Standard Class** offered the best balance between shipment volume, transportation cost, and lead time.

This analysis supports strategic decision-making by helping logistics managers balance service quality with transportation expenses and identify the most cost-effective shipping options.

6. Route Drill-Down

The **Route Drill-Down** section provides detailed operational insights for individual Factory-to-Customer routes. Users can select any route to examine shipment count, average lead time, transportation cost, gross profit, lead time variability, shipment distribution, and detailed shipment records.

For example, the selected route **Lot's O' Nuts → Alabama** recorded:

- Total Shipments: **34**
- Average Lead Time: **1295.21 days**
- Average Transportation Cost: **\$4.37**
- Average Gross Profit: **\$9.62**
- Lead Time Variability: **253.07 days**

The accompanying lead time distribution histogram and ship mode distribution chart further illustrate shipment characteristics for the selected route. This drill-down functionality enables operational managers to identify route-specific inefficiencies and prioritize improvement initiatives at a granular level.

7. State-Level Performance

The **State-Level Performance** section summarizes shipping performance for each destination state by reporting total shipment volume and average shipping lead time.

The visualization enables comparison of logistics performance across different geographic locations and helps identify states experiencing consistently longer shipment durations. States such as **West Virginia, North Dakota, and Saskatchewan** recorded comparatively higher average lead times, whereas other regions demonstrated more favorable delivery performance.

This state-level analysis assists logistics planners in identifying geographically concentrated delivery challenges and supports data-driven decisions regarding warehouse allocation, transportation routing, and carrier management.

8. Recent Shipment Timeline

The **Recent Shipment Timeline** analyzes monthly shipment activity by displaying trends in both order volume and average shipping lead time throughout the analysis period.

The dashboard indicates that shipment volume increased substantially during late 2025, with the **highest monthly order volume occurring in December 2025 (848 orders)**. The timeline

also reveals that the **fastest average delivery performance occurred in January 2024**, when the average lead time reached **1067.16 days**.

By monitoring both shipment demand and delivery performance simultaneously, this visualization enables organizations to detect seasonal demand patterns, evaluate operational stability, and anticipate future logistics capacity requirements.

9. Interactive Filtered Dataset

The **Interactive Filtered Dataset** provides users with direct access to all shipment records matching the currently selected dashboard filters. Users can refine the dataset by **date range, region, state, ship mode, and maximum lead time**, with all dashboard visualizations updating dynamically based on the selected criteria.

The dashboard also includes an option to download the filtered data in CSV format, enabling additional analysis using Microsoft Excel, Power BI, or other analytical tools. This interactive capability supports exploratory data analysis, operational reporting, and informed decision-making while ensuring consistency across all dashboard KPIs and visualizations.

Overall Dashboard Summary

The **Factory-to-Customer Shipping Route Efficiency Dashboard** provides a comprehensive view of the logistics performance of Nassau Candy Distributor by integrating KPI summary cards, geographic analysis, regional bottleneck identification, shipping mode comparison, cost-versus-performance evaluation, route-level exploration, state-wise performance monitoring, shipment timeline analysis, and an interactive filtered dataset. Together, these components enable stakeholders to monitor operational efficiency, identify logistics bottlenecks, evaluate transportation costs, and make data-driven decisions to optimize factory-to-customer distribution performance.

9. Conclusion

This project successfully developed a **Factory-to-Customer Shipping Route Efficiency Dashboard** for the Nassau Candy Distributor to evaluate logistics performance across its distribution network. By integrating data preprocessing, feature engineering, exploratory data analysis, and interactive dashboard development using **Python, Pandas, Plotly, and Streamlit**, the project transformed raw shipment data into meaningful business insights that support operational decision-making.

The dashboard analyzed **10,194 shipment records** across **196 Factory-to-Customer routes** serving **59 states and provinces**. A comprehensive set of Key Performance Indicators (KPIs) was developed, including Shipping Lead Time, Route Volume, Delay Frequency, Route Efficiency Score, Transportation Cost, and Gross Profit. These KPIs enabled a systematic assessment of shipping performance from multiple operational perspectives.

The route analysis revealed considerable variation in shipping performance across factory-to-customer routes. Several routes consistently recorded lower average lead times and higher efficiency scores, while others experienced significantly longer shipment durations, highlighting opportunities for logistics optimization. The Geographic Shipping Heat Map and State-Level Performance analysis further demonstrated that shipping efficiency varied geographically, enabling the identification of states requiring operational attention.

The Regional Bottleneck Analysis identified the **Pacific** and **Atlantic** regions as the most critical operational bottlenecks because of their high shipment volumes and elevated bottleneck scores. In contrast, the **Gulf** region demonstrated comparatively better logistics performance, while the **Interior** region exhibited longer shipment durations despite handling fewer shipments. These findings provide valuable guidance for prioritizing future logistics improvement initiatives.

The Ship Mode Comparison showed that **Standard Class** handled the largest proportion of shipments while maintaining the lowest average lead time within the dataset, indicating consistent operational performance. The Cost vs. Lead Time Analysis further demonstrated that transportation cost alone was not a reliable predictor of delivery performance, emphasizing the importance of balancing cost, shipment volume, and operational efficiency when selecting shipping strategies.

The interactive Route Drill-Down functionality enabled detailed evaluation of individual factory-to-customer routes by combining shipment counts, lead time distributions, transportation costs, gross profit, and ship mode utilization. The inclusion of dynamic filters and downloadable datasets enhanced the dashboard's usability by allowing users to perform customized analyses based on specific regions, states, shipping modes, and time periods.

Overall, the project demonstrates how interactive business intelligence dashboards can transform logistics data into actionable insights that support evidence-based decision-making. The developed dashboard provides logistics managers with an effective tool for monitoring shipping performance, identifying bottlenecks, evaluating transportation strategies, and prioritizing operational improvements across the distribution network.

Although the analysis provides valuable operational insights, the findings should be interpreted within the context of the available dataset. The lead-time values were calculated directly from the original **Order Date** and **Ship Date** fields and were preserved to maintain data integrity. Consequently, the results are intended to compare relative shipping performance within the dataset rather than represent actual real-world delivery durations.

The project demonstrates the practical application of **data analytics, visualization, and dashboard development** in solving logistics and supply chain challenges and illustrates how modern analytical tools can support continuous performance monitoring and strategic decision-making in distribution operations.

#####THE END#####